OXFORD UNIVERSITY PRESS

Predictors of Cigarette Smoking, Smokeless Tobacco Consumption, and Use of both forms in Adolescents in South Asia

Author(s): Masuma Pervin Mishu, Faraz Siddiqui, Radha Shukla, Mona Kanaan, Omara Dogar and Kamran Siddiqi

Source: *Nicotine & Tobacco Research*, June 2021, Vol. 23, No. 6 (June 2021), pp. 956-965

Published by: Oxford University Press

Stable URL: https://www.jstor.org/stable/10.2307/27397203

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Oxford University Press is collaborating with JSTOR to digitize, preserve and extend access to *Nicotine & Tobacco Research*

JSTOR

This content downloaded from 103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

Nicotine & Tobacco Research, 2021, 956–965
doi:10.1093/ntr/ntaa202
Original Investigation
Received May 4, 2020; Editorial Decision September 24, 2020; Accepted September 29, 2020

Original Investigation

Predictors of Cigarette Smoking, Smokeless Tobacco Consumption, and Use of both forms in Adolescents in South Asia: A Secondary Analysis of the Global Youth Tobacco Surveys

Masuma Pervin Mishu PhD[1], Faraz Siddiqui MSc[1], Radha Shukla MBBS[1], Mona Kanaan PhD[1], Omara Dogar PhD[1,2], Kamran Siddiqi PhD[1,3]

[1]Department of Health Science, University of York, York, UK; [2]Usher Institute, University of Edinburgh, Edinburgh, UK; [3]Centre for Health and Population Sciences, Hull York Medical School, University of York, York, UK

Corresponding Author: Masuma Pervin Mishu, PhD, Department of Health Sciences, University of York, Heslington, York YO10 5DD, UK. Telephone: (01904) 32 1772; Fax: (01904) 32 1383; E-mail: masuma.mishu@york.ac.uk

Abstract

Introduction: Tobacco uptake in adolescents is associated with a range of predictors. We examined the predictors of cigarette smoking, smokeless tobacco (ST) consumption, and use of both ST and cigarettes among adolescents in four South Asian countries.

Methods: We analyzed the Global Youth Tobacco Surveys (GYTS) data for Bangladesh (2013), India (2009), Pakistan (2013), and Sri Lanka (2015), using multinomial regression to examine associations between several predictors and tobacco use.

Results: Data from 23 681 adolescents were analyzed. Overall, 82.8% of the study population were between 13 and 15 years and 52.7% were girls, 2% were cigarette smokers, 6.5% were ST users, and 1.1% used both ST and cigarettes, in the past 30 days. Exposure to smoking in public places was associated with past 30-day smoking (relative risk ratio [RRR] 5.59, 95% confidence intervals [CI] 4.28–7.28), ST use (RRR 2.07, 95% CI 1.84–2.32), and use of both ST and cigarettes (RRR 11.42, 95% CI 7.44–17.54). Exposure to tobacco use in electronic media and being offered free tobacco products were associated with all forms of tobacco use. Shopkeepers' refusal to sell cigarettes protected adolescents from smoking (RRR 0.47, 95% CI 0.36–0.63) and ST use (RRR 0.65, 95% CI 0.45–0.95). However, exposure to antitobacco mass media messages was not protective for any form of tobacco use. Adolescents taught at school about harmful effects of tobacco were less likely to use ST; no evidence of this association was observed for smoking.

Conclusion: The associations between tobacco use and protobacco factors were strong, but the associations with antitobacco factors lacked strength and consistency in this study population.

Implications: The predictors of adolescents using different tobacco products, crucial to inform and evaluate tobacco control efforts, are poorly understood. We investigated the associations between several environmental-level factors and cigarette smoking, ST consumption, and use of both forms among adolescents, whereas most of the previous studies focused on individual-level factors. Our study found strong associations between tobacco use and protobacco factors and lack of strength and consistency in associations between antitobacco factors and tobacco use in the study population. Our results indicate that the current tobacco control policies need strengthening to curb the tobacco epidemic in these countries.

© The Author(s) 2020. Published by Oxford University Press on behalf of the Society for Research on Nicotine and Tobacco. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com.

This content downloaded from 103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

Introduction

Tobacco use among youth is a global public health challenge.[1] In 2016, an estimated 13.6% adolescents aged 12–15 years were current tobacco users in 68 low- and middle-income countries.[2] Most smokers start smoking in adolescence,[1] and among those adolescents who experiment with smoking, between one-third and one-half become regular smokers.[3] In fact, youth are highly vulnerable to tobacco industry marketing[4,5] and hence targeted.[6] Tobacco use is common among adolescents in South Asia.[7] The prevalence of current tobacco use in 13–15 year olds in Bangladesh, India, Pakistan, and Sri Lanka is 9.2%, 14.6%, 10.7%, and 3.7%, respectively.[8] Smokeless tobacco (ST) use is a particular concern,[9] with its prevalence being 4.5%, 9.0%, 5.3%, and 2.4% in the four aforementioned countries, respectively.[8]

The high prevalence of adolescent tobacco use in South Asia can be attributed to several factors. Tobacco products are diverse and encompass smoking and smokeless forms.[10] In contrast, tobacco control regulations are considerably weak.[11] Cultural acceptability, high level of addictiveness,[12] easy accessibility, low cost,[13] and lack of prohibitive legislation and limited awareness of health effects are the main factors encouraging ST use in the South Asian population.[14]

Tobacco uptake among youth is influenced by several anti- and protobacco factors. There is sufficient evidence on the predictors of smoking among youth, primarily in high-income countries; these include sociodemographic, personal/psychological, social, and environmental factors.[15] Protobacco marketing[16] and exposure to cigarette advertising and promotion[17] increase tobacco use among adolescents; these include direct-to-consumer strategies like receiving mail from tobacco companies and pop-ups while browsing Internet.[18] Moreover, exposure to smoking in public places and mass media (protobacco depictions in films, television, and videos) are considered to increase the risk for smoking uptake in youth.[16,19] On the other hand, factors such as bans on tobacco point-of-sale display and mass media campaigns are considered protective.[20,21] The evidence on the effectiveness of school-based educational programs in preventing smoking uptake is mixed.[22] Only a handful of studies from South Asia have so far identified anti- and protobacco factors (for instance, age, gender, family and peer influence, exposure to smoking at home and public places, availability of tobacco products, teaching of smoking hazards in schools), influencing uptake of smoking,[23,24] ST consumption,[25] and tobacco use in general[26,27] among youth.

The evidence supporting the World Health Organization (WHO)–Framework Convention on Tobacco Control (FCTC), while applicable to all forms of tobacco, is mostly derived from research conducted on smoking and mainly from high-income countries. Given that, country-level policies formulated on the grounds of such evidence may have limited transferability to low-income settings, particularly where ST and use of both tobacco products are major problems. A detailed analysis of the predictors of tobacco use behavior (smoking, ST, and use of both ST and cigarettes) is particularly important to inform tobacco control policies. Most of the previous studies in South Asia focused on individual-level knowledge and attitudes and family and peer influences on smoking or on tobacco use, in general. Very limited studies have investigated predictors of cigarette smoking, ST consumption, and use of both ST and cigarettes as separate behaviors among adolescents and considered wider environmental predictors.

Our study aims to test the associations between several pro- and antitobacco predictors and smoking, ST consumption, and concurrent use of both forms among adolescents. We hypothesized that the antitobacco factors (refused to purchase cigarettes due to age restrictions, exposure to antitobacco mass media messages, exposure to antitobacco messages at community events, and taught in class about the dangers of tobacco use, health warning on cigarette pack) would be negatively associated with tobacco use, whereas protobacco factors (exposed to tobacco use in electronic media, offered free tobacco products, and exposure to cigarettes at any public places, advertisement at point of sell, exposure at school premises) would be positively associated with tobacco use (Supplementary Figure 1).

Methods

We conducted a secondary analysis of the Global Youth Tobacco Surveys (GYTS) datasets for Bangladesh (2013), India (2009), Pakistan (2013), and Sri Lanka (2015), accessible freely from the US Centre for Disease Control and Prevention (CDC) website.[28] GYTS is a cross sectional survey, designed by the WHO and CDC for measuring tobacco prevalence in school going children (aged between 11 and 19 years), conducted in 151 countries.[29] GYTS is a school-based survey that employed a two-stage cluster sampling design. GYTS questionnaires were paper-based bubble sheets that were self-administered in classrooms and anonymized to ensure confidentiality. Survey methods and procedures are detailed elsewhere.[30–33]

Obtaining the Analytical Sample

A total of 3245 students from Bangladesh, 11 768 from India,[34] 8723 from Pakistan, and 1505 from Sri Lanka took part in GYTS. Data for 24 students from India, 47 from Pakistan, and 2 from Sri Lanka were dropped from the analyses due to missing outcome, and these were not different from the rest of the study participants. The total number of students surveyed with response rates and numbers analyzed was presented in Figure 1.

The GYTS questionnaire included the following topics: tobacco use (smoking and ST), cessation, secondhand smoke, pro- and antitobacco media messages and advertising, access to and availability of tobacco products, and knowledge and attitudes toward tobacco. The following variables were available for the four countries and were used in the analysis.

The outcome variable was derived from "any cigarette smoking and/or ST use in the past 30 days." To obtain this variable, the responses to the following questions were merged: "During the past 30 days, on how many days did you smoke cigarettes?" and "During the past 30 days, on how many days did you use smokeless tobacco?" The response options for these questions were as follows: 0, 1 or 2, 3–5, 6–9, 10–19, 20–29, or all 30 days (with only exception for ST use in Pakistan: "During the past 30 days, did you use any form of smokeless tobacco products?"). The response options were dichotomized for the respective questions into no use (0 days) and any use (≥1 day), before merging the questions. The derived outcome variable after merging the smoking and ST questions had four categories: (1) neither smoked cigarette nor used any ST in the past 30 days, (2) exclusive cigarette smoking in the past 30 days, (3) exclusive ST use in the past 30 days, and (4) use of both ST and cigarettes in the past 30 days.

Explanatory variables were hypothesized as antitobacco and protobacco as follows:

Antitobacco variables: (1) "refused to purchase cigarettes due to age restrictions" (question in survey: "During the past 30 days, did

This content downloaded from
103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

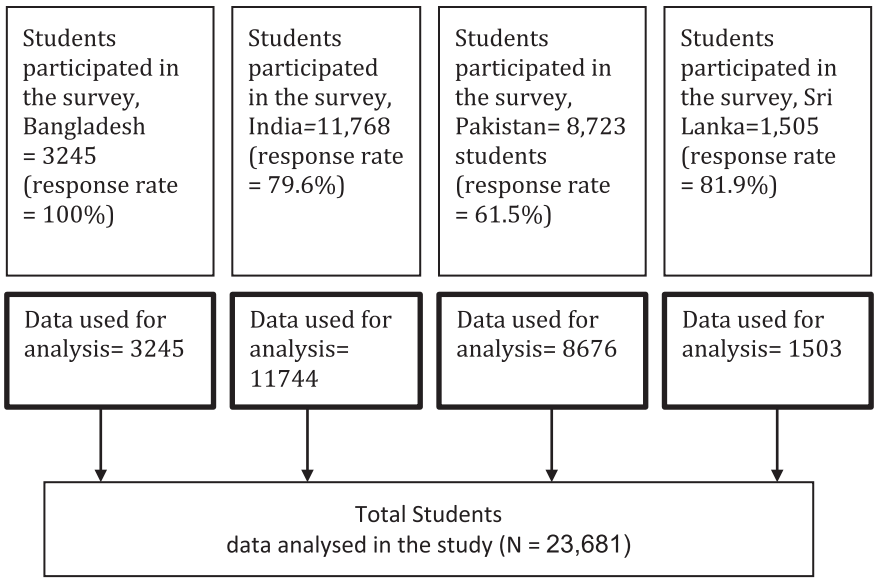


Figure 1. Number of study participants in Global Youth Tobacco Surveys from each country and data analyzed.

anyone refuse to sell you cigarettes because of your age?" with response options: I did not try to buy; yes, someone refused to sell me cigarettes because of my age; and no, my age did not keep me from buying cigarettes). (2) "Exposure to antitobacco mass media messages" (question in survey: "During the past 30 days, did you see or hear any antitobacco media messages on TV, radio, internet, billboards, posters, newspapers, magazines or movies?" with response options: yes and no). (3) "Exposure to antitobacco messages at community events" (question in survey: "During the past 30 days, did you see or hear any antitobacco media messages at sports events, fairs, concerts, or community events, or social gatherings?" with response options: did not go to sports events, fairs, concerts, or community events, or social gatherings in the past 30 days; yes; and no). (4) "Taught in class about the dangers of tobacco use" (question in survey: "During the past 12 months, were you taught in any of your classes about the dangers of tobacco use?" with response options: yes; no; and don't know).

Protobacco variables: (1) "exposed to tobacco use in electronic media" (question in survey: "During the past 30 days, did you see any people using tobacco when you watched TV, videos, or movies?" with response options: did not watch TV, videos, or movies in the past 30 days; yes; and no). (2) "Offered free tobacco products" (question in survey: "Has a person working for a tobacco company ever offered you a free tobacco product?" with response options: yes and no. (3) "Exposure to cigarettes at any public places" was used for analysis. We derived this variable from the following two questions: "During the past 7 days, on how many days has anyone smoked in your presence, inside any enclosed public place, other than your home?" and "During the past 7 days, on how many days has anyone smoked in your presence, at any outdoor public places?" The response options for both questions were as follows: 0, 1–2, 3–4, 5–6, or 7 days. These two variables were merged to create the new variable "any exposure in public places," which was defined as no exposure if the response was 0 days for both original variables otherwise they were considered to be exposed.

Sociodemographic variables: age ("How old are you?" with response options, 11 years old or younger, 12, 13, 14, 15, 16, 17 years old or older), sex ("What is your sex?" with response options, male and female), and country were considered as confounding factors. GYTS data does not provide any socioeconomic information.

Three additional factors available for all countries except India were used in an additional analysis. (1) "Health warning on cigarette pack" (question in survey: "During the past 30 days, did you see any health warning on cigarette Packages?" with response options: yes, I did not think much of them; yes, it led me to think about quitting smoking or not starting smoking; no). (2) "Seen any advertisements or promotions for tobacco products at points of sale" (question in survey: "During the past 30 days, did you see any advertisements or promotions for tobacco products at points of sale [stores, shops, street vendor])?" with response options: did not visit any points of sale in the past 30 days; yes; and no). (3) "Exposure to cigarette at school premises" (question in survey: "During the past 30 days, did you see anyone smoke inside the school building or outside on school property?" with response options: yes and no).

Statistical Analysis

Absolute and relative frequencies were provided for demographic variables (country, age, sex) and for the distribution of anti- and protobacco predictors by the type of tobacco used. Pearson's chi-square tests were used to detect bivariate associations between the covariates (age, sex, country, anti- and protobacco predictors) and tobacco use. Outcome "past 30 days any tobacco use" was analyzed using multinomial logistic regression (with "neither smoked cigarette nor used any ST in past 30 days" as the reference category), adjusted for explanatory variables: refused purchase of cigarettes due to age restrictions, antitobacco media messages on mass media, antitobacco messages on community events, and taught in class about the dangers of tobacco use, cigarettes smoking exposure, offered free tobacco products, watching people using tobacco in TV and likely confounders: age, sex, and country. The relative risk ratio (RRR) and 95% confidence intervals (CI) were reported.

Additional analysis was run for the combined data of Bangladesh, Pakistan, and Sri Lanka controlling for the following additional variables: health warning on pack, advertisements at points of sale, and exposure to cigarettes at school premises. The results of the

This content downloaded from 103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

additional analysis were compared to those of the primary model excluding India.

All analyses were performed using STATA (version 16).[35] Statistical significance level was set at .05 a priori. We did not account for the survey design as we were interested in the associations only and not nationally representative estimates. As the proportion of missing data on our variables of interest was low (<10%), all statistical analyses were conducted on complete cases only.

Results

Data from a sample of 23 681 adolescents collected between 2009 and 2015 were analyzed. Overall, 13.1% of the total sample were from Bangladesh, 47.8% from India, 32.9% from Pakistan, and 6.3% from Sri Lanka. Overall, 82.8% of the study population were between 13 and 15 years and 52.7% were girls. Country wise, the highest percentage of exclusive smoking in the past 30 day was reported in India (2.4%), followed by Pakistan (1.8%), Bangladesh (1.3%), and Sri Lanka (0.7%). On the other hand, the highest percentage of exclusive ST use in the past 30 day was reported in Sri Lanka (27.2%), followed by India (6.1%), Pakistan (4.1%), and Bangladesh (3.6%) (Supplementary Table 1). Exposure to pro- and antitobacco factors by countries is presented in Supplementary Tables 2–4.

The overall self-reported use of any tobacco (in the past 30 days) in our sample (smoking, ST, or both) was 9.5% (exclusive smoking 2.0%, exclusive ST use 6.4%, and use of both ST and cigarettes in the past 30 day 1.1%) (Table 1). Tobacco use was more common in males (54.1%) than in females (45.9%). Among antitobacco factors, 4.4% participants in our sample were refused purchase of cigarettes by the sellers due to age restrictions. Fifty-nine percent reported being taught in class about dangers of tobacco, whereas 70.1% and 34.9% reported being exposed to antitobacco messages via mass media and at community events, respectively. Among protobacco factors, highest reported exposure was for tobacco use in electronic media (64.1%), followed by cigarette smoking in public places (42.4%), and offers of free tobacco products (9.5%) (Table 1).

The results of adjusted multinomial regression on the associations between anti- and protobacco factors with tobacco use showed that those who were refused purchase of cigarettes due to age restrictions were less likely to be smokers (RRR 0.47, 95% CI 0.36–0.63) or ST users (RRR 0.65, 95% CI 0.45–0.95), compared with those who were not refused at the point of sale. However, this association was not observed among users of both ST and cigarettes in the past 30 days. No associations were observed between exposure to antitobacco messages in the mass media and any form of tobacco use. Those exposed to antitobacco messages at community events were more likely to be exclusive smokers (RRR 1.59, 95% CI 1.15–2.20), exclusive ST users (RRR 1.80, 95% CI 1.54–2.11), or using both forms (RRR 2.20, 95% CI 1.41–3.44) compared with those not exposed. Those who were taught about tobacco-related harms in schools were less likely to be ST users (RRR 0.64, 95% CI 0.56–0.76) than those who were not taught; however, no association was observed with smoking or use of both ST and cigarettes (Table 2).

Among protobacco predictors, tobacco use in electronic media, offer of free tobacco products and exposure to smoking at public places increased the risk of all forms of tobacco use. Those exposed to tobacco use in electronic media were twice as likely to be exclusive smoker (RRR 2.14, 95% CI 1.39–3.31), 1.62 times to be ST user (RRR 1.62, 95% CI 1.33–1.96), and 2.62 times to be users of both forms (RRR 2.62, 95% CI 1.31–5.26) than those not exposed. Those offered a free tobacco product were also more likely to be exclusive smokers (RRR 1.45, 95% CI 1.05–2.00), ST users (RRR 1.48, 95% CI 1.23–1.78), and users of both forms (RRR 1.99, 95% CI 1.34–2.94) than those who were not offered. Among protobacco factors, the strongest associations were observed for exposure to cigarette smoking in public places. Those being exposed were nearly six times more likely to be smokers (RRR 5.59, 95% CI 4.28–7.28), twice as likely to be ST users (RRR 2.07, 95% CI 1.84–2.32), and 11 times more likely to be users of both forms (RRR 11.42, 95% CI 7.44–17.54) than those who were not exposed (Table 2).

Additional Analysis

The restricted analyses of Bangladesh, Sri Lanka, and Pakistan data prior to adding any additional variables showed that the estimates for smoking and use of both forms were almost similar to the main analysis. While for ST use, no association observed for "taught in the class on dangers of tobacco," which was the case in the main analysis (Table 3).

The adjusted estimates of the additional three variables (health warning on pack, advertisements at points of sale and exposure to cigarette at school premises) showed that exposure to cigarette smoking within school premises was positively associated with any form of tobacco use. However, those exposed to "health warnings on cigarette packs" were more likely to be exclusive smokers (RRR 2.00, 95% CI 1.26–3.17) or users of both ST and cigarettes (RRR 4.16, 95% CI 1.75–9.92) than those who were not exposed. The associations of other variables with smoking and use of both forms were generally similar to the main model (Supplementary Table 5).

Discussion

Our findings suggest that ST was the most common form of tobacco use in this study population. Tobacco use in adolescents was positively and strongly associated with all hypothesized protobacco predictors, that is, exposure to tobacco use in electronic media, offer of free tobacco products, and exposure to smoking in public places. Among antitobacco predictors, only shopkeepers' refusal to sell cigarettes to adolescents were protective against smoking and ST use. Being taught at school about the harmful effects of tobacco was only partially protective, that is, from ST use but not from cigarettes. Exposure to antitobacco mass media messages was not protective from any form of tobacco use. Furthermore, exposure to antitobacco messages at community events was positively associated with tobacco use. It is possible that those using tobacco may be more sensitized towards reporting exposure to antitobacco messages at the community events. A positive association was also observed between exposure to "health warnings on cigarette packs" with smoking and use of both forms, which is plausible as cigarette users are likely to be more exposed to health warnings than nonusers. For ST use, a lack of association was not unexpected as most of the ST products in South Asia are sold without health warnings.[36]

Our study found that the proportion of tobacco use increased markedly between ages 12–13 years and age 14–15 years, and among boys, which was similar to the findings by Xi et al. based on data from 68 low- and middle-income countries.[2] GYTS data from 1999 to 2005 in 131 countries showed that 54.2% adolescents were exposed to smoking in public places.[37] This figure was 42.4% in our study. Similar to our study, Xi et al. found that secondhand smoke exposure (55.9%)

This content downloaded from
103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

Table 1. Bivariate Analyses of Distribution of Tobacco Use (Exclusive Smoking, Exclusive ST, and Use of Both Forms) by Demographic Characteristics, Pro- and Antitobacco Factors: Data From Bangladesh, India, Pakistan, and Sri Lanka (Presented as Count and %; *N* max = 23 681)

	Total, *N* (%)		Nonuser		Exclusive smoker		Exclusive smokeless tobacco		User of both forms		*p* (χ^2 test)
	21 430 (90.49)				469 (1.98)		1527 (6.45)		255 (1.08)		
	n	%	*n*	%	*n*	%	*n*	%	*n*	%	
Country											
Bangladesh	3092	13.06	2929	13.67	41	8.74	112	7.33	10	3.92	<.001
India	11 318	47.79	10 178	47.49	277	59.06	690	45.19	173	67.84	
Pakistan	7784	32.87	7265	33.90	140	29.85	320	20.96	59	23.14	
Sri Lanka	1487	6.28	1058	4.94	11	2.35	405	26.52	13	5.10	
Sex											
Male	11 102	47.31	9742	45.86	339	73.38	850	56.14	171	68.67	<.001
Female	12 365	52.69	11 500	54.14	123	26.62	664	43.86	78	31.33	
Age											
≤11 to 12 years	2069	8.79	1940	9.11	26	5.60	87	5.72	16	6.40	<.001
13 to 15 years	19 496	82.83	17 606	82.65	376	81.03	1314	86.39	200	80.00	
16 to ≥17 years	1973	8.38	1757	8.25	62	13.36	120	7.89	34	13.60	
Refused to purchase cigarettes[a]											
Did not try to buy	21 558	91.61	20 037	93.97	109	24.17	1365	90.04	47	19.34	<.001
Yes	1045	4.44	728	3.41	137	30.38	75	4.95	105	43.21	
No	929	3.95	557	2.61	205	45.45	76	5.01	91	37.45	
Exposure to antitobacco mass media messages[a]											
Yes	16 367	70.09	14 812	70.03	330	72.53	1056	70.45	169	68.15	.601
No	6986	29.91	6339	29.97	125	27.47	443	29.55	79	31.85	
Exposure to antitobacco messages at community events[a]											
I did not go to any event[a]	8204	34.99	7729	36.39	86	18.70	348	23.18	41	16.67	<.001
Yes	10 114	43.14	8845	41.65	280	60.87	823	54.83	166	67.48	
No	5127	21.87	4664	21.96	94	20.43	330	21.99	39	15.85	
Taught in class on dangers of tobacco in past 12 mo											
Yes	13 787	58.92	12 615	59.54	263	57.42	757	50.20	152	61.54	<.001
No	6705	28.66	6049	28.55	139	30.35	446	29.58	71	28.74	
I don't know	2907	12.42	2522	11.90	56	12.23	305	20.23	24	9.72	
Exposed to tobacco use in electronic media[a]											
Did not watch TV[a]	4909	21.01	4488	21.22	88	19.01	266	17.65	67	27.24	<.001
Yes	14 986	64.15	13 427	63.50	334	72.14	1066	70.74	159	64.63	
No	3467	14.84	3231	15.28	41	8.86	175	11.61	20	8.13	
Offered free tobacco product ever											
Yes	2207	9.54	1908	9.10	77	17.00	172	11.64	50	20.75	<.001
No	20 933	90.46	19 060	90.90	376	83.00	1306	88.36	191	79.25	
Exposure to cigarette at any public places											
Yes	10 030	42.44	8574	40.09	369	78.85	873	57.32	214	84.58	<.001
No	13 601	57.56	12 813	59.91	99	21.15	650	42.68	39	15.42	

[a]Past 30 day.

This content downloaded from 103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

Table 2. Result of Adjusted[a] Multinomial Logistic Regressions on Association Between Tobacco Use (Exclusive Smoking, Exclusive ST, and Dual Use) (Ref Category: Nonsmoker and Non-ST User) and Related Covariates: Data from Bangladesh, India, Pakistan, and Sri Lanka (*N* = 22 007)

	Exclusive smoking			Exclusive ST use			Use of both forms		
	RRR	95% CI	*p*	RRR	95% CI	*p*	RRR	95% CI	*p*
Antitobacco factors									
Refused to purchase cigarettes (past 30 day) (ref: no)									
Did not try to buy	0.01	0.01, 0.02	<.001	0.46	0.35, 0.60	<.001	0.01	0.01, 0.02	<.001
Yes	0.47	0.36, 0.63	<.001	0.65	0.45, 0.95	.026	0.86	0.61, 1.23	.416
Exposure to antitobacco mass media messages (past 30 day) (ref: no)									
Yes	1.18	0.90, 1.55	.229	0.96	0.83, 1.10	.541	0.94	0.66, 1.33	.735
Exposure to antitobacco messages at community events (past 30 day) (ref: no)									
I did not go to any event in past month	0.71	0.49, 1.03	.069	0.58	0.49, 0.69	<.001	0.49	0.29, 0.86	.012
Yes	1.59	1.15, 2.20	.005	1.80	1.54, 2.11	<.001	2.20	1.41, 3.44	.001
Taught in class on dangers of tobacco in past 12 month (ref: no)									
Yes	0.95	0.73, 1.24	.712	0.64	0.56, 0.76	<.001	1.11	0.78, 1.57	.556
I don't know	1.11	0.76, 1.63	.577	1.55	1.31, 1.83	<.001	0.73	0.40, 1.35	.319
Tobacco use motivating factors									
Exposed to tobacco use in electronic media in past 30 day (ref: No)									
Did not watch TV in last 30 day	1.76	1.08, 2.88	.023	1.41	1.13, 1.76	.003	4.34	2.09, 9.00	<.001
Yes	2.14	1.39, 3.31	.001	1.62	1.33, 1.96	<.001	2.62	1.31, 5.26	.007
Offered free tobacco product ever (ref: no)									
Yes	1.45	1.05, 2.00	.022	1.48	1.23, 1.78	<.001	1.99	1.34, 2.94	.001
Exposure to cigarette at any public places (ref: no)									
Yes	5.59	4.28, 7.28	<.001	2.07	1.84, 2.32	<.001	11.42	7.44, 17.54	<.001
Other covariates									
Sex (ref: male)									
Female	0.55	0.43, 0.71	<.001	0.73	0.65, 0.82	<.001	0.61	0.44, 0.85	.003
Age (ref: ≤11 to 12 years)									
13 to 15 y old	1.75	1.04, 2.94	.034	2.24	1.72, 2.93	<.001	1.17	0.64, 2.13	.617
16 to ≥17 y old	2.30	1.27, 4.17	.006	1.60	1.15, 2.21	.005	1.36	0.66, 2.82	.405

CI = confidence interval; RRR = relative risk ratio; ST = smokeless tobacco.

[a]Adjusted for country, age, and sex.

This content downloaded from 103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

Table 3. Association Between Tobacco Use (Exclusive Smoking, Exclusive ST, and Use of Both Forms) (Ref Category: Nonsmoker and Non-ST User) and Related Covariates (Result of Adjusted Analyses Multinomial Logistic Regression Data From Bangladesh, Pakistan, and Sri Lanka (Without the Additional Variables) (N = 11 441)

	Exclusive smoking			Exclusive ST use			Use of both forms		
	RRR	95% CI	*p*	RRR	95% CI	*p*	RRR	95% CI	*p*
Antitobacco factors									
Refused to purchase cigarettes (past 30 day) (ref: no)									
Did not try to buy	0.05	0.03, 0.08	<.001	0.24	0.17, 0.34	<.001	0.02	0.01, 0.05	<.001
Yes	0.86	0.53, 1.40	.549	0.37	0.23, 0.60	<.001	0.33	0.16, 0.68	.003
Exposure to antitobacco mass media messages (past 30 day) (ref = no)									
Yes	1.20	0.83, 1.75	.331	1.51	1.25, 1.81	<.001	1.52	0.85, 2.73	.160
Exposure to antitobacco messages at community events (past 30 day) (ref: no)									
I did not go to any event in past month	0.55	0.36, 0.83	.005	0.70	0.57, 0.87	.001	0.39	0.19, 0.80	.011
Yes	0.82	0.53, 1.29	.398	2.42	1.97, 2.96	<.001	1.92	1.04, 3.53	.037
Taught in class on dangers of tobacco in past 12 month (ref = no)									
Yes	1.30	0.87, 1.95	.202	0.90	0.75, 1.09	.281	1.37	0.74, 2.52	.316
I don't know	1.64	0.96, 2.79	.070	1.35	1.05, 1.74	.017	0.93	0.37, 2.37	.886
Tobacco use motivating factors									
Exposed to tobacco use in electronic media in past 30 day (ref: no)									
Did not watch TV in last 30 day	1.50	0.80, 2.80	.204	1.31	1.01, 1.68	.037	1.05	0.41, 2.72	.914
Yes	1.97	1.14, 3.41	.015	1.27	1.01, 1.59	.041	1.49	0.67, 3.32	.327
Offered free tobacco product ever (ref: no)									
Yes	2.08	1.28, 3.38	.003	1.72	1.30, 2.28	<.001	3.88	2.08, 7.25	<.001
Exposure to cigarette at any public places (ref: no)									
Yes	3.04	1.99, 4.63	.000	1.24	1.06, 1.46	.008	5.02	2.44, 10.32	<.001
Other covariates									
Sex (ref: male)									
Female	0.27	0.17, 0.43	.000	0.74	0.63, 0.87	<.001	0.17	0.07, 0.39	<.001
Age (ref: ≤11 to 12 years)									
13 to 15 y old	1.50	0.69, 3.22	.302	2.23	1.58, 3.14	<.001	1.44	0.43, 4.88	.552
16 to ≥17 y old	3.71	1.66, 8.30	.001	2.12	1.43, 3.15	<.001	3.09	0.86, 11.02	.082

CI = confidence interval; RRR = relative risk ratio; ST = smokeless tobacco.

This content downloaded from 103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

(exposure at any public places and exposure at home and parental tobacco use) was a risk factor for adolescent's tobacco use (OR 2.56, 95% CI 2.43–2.69).[2] However, these two studies reported weighted prevalence where as our study reported the proportion in the study sample.

Consistent with our findings, one study in Karachi showed that access to free tobacco products influenced tobacco-related behavior among adolescents.[25] Easy or sometimes free access to tobacco products and lack of restrictions on sales to minors increased the possibility of tobacco use among youths.[27]

Our finding of a strong association between exposure to tobacco in media and tobacco use among adolescents was similar to the evidence from Republic of Congo GYTS-2006 study reporting that those exposed to tobacco advertisement on TV, billboards, and in newspapers/magazines were twice as likely to be current ST users than those who were not exposed.[38] Another study reported that a reduction in mass media exposure to tobacco products decreases the appeal of smoking among adolescents.[39] A Cochrane systematic review has also confirmed that nonsmoking adolescents who were more aware of or receptive to tobacco advertising were more likely to become smokers later and cigarette advertising and promotion increased smoking in youth.[17]

Similar to our study, antitobacco media messages showed no association with tobacco use and areca nut use in a study in Mumbai.[40] A possible explanation, as noted in this study, is that the antitobacco messages, because they are not specifically designed for targeting younger audiences, might have a reduced impact and inadvertently draw attention to tobacco. Our study also did not observe any protective effect of exposure of health warning on cigarette packs among this age group. This was supported by other studies that reported pictorial health warning on pack do not increase youth's expectations to be nonsmokers a year later[41] and have no effect on beliefs about cancer or addiction among nonsmoking adolescent boys.[42]

The previous round of GYTS data from Bangladesh, Sri Lanka, and Nepal reported that class lessons on the dangers of smoking was associated with reduced tobacco usage among adolescents.[27] However, we found no evidence of association with teaching harms of tobacco in school with smoking or use of both forms in this study population, which is in line with the mixed evidence-base on the effect of school-based educational programs on youth smoking uptake.[22]

Limitations and Strengths

Our study has some limitations that deserve mentioning. First, we did not account for survey weights in our analysis as we were interested in the associations and not nationally representative estimates. However, the prevalence of current cigarette smoking and ST use as reported in the study population was broadly similar to the nationally representative prevalence data reported by the countries (only exception of past 30-day ST use in Sri Lanka. See Supplementary Table 6 for details). Second, the GYTS data are self-reported, which might not necessarily reflect the true prevalence. Being cross sectional data, an analysis based on GYTS cannot infer causality. Besides, reverse causality could be an issue, which might explain the observed positive association between tobacco use and exposure of "anti tobacco messages at community events" and "health warning on cigarette packs" as the users might report the exposure more than the non users. Third, we merged data from surveys that collected information in different years (2009, 2013, and 2015); therefore, comparison of proportions between countries should be made with caution. In addition, surveys were done before 2010 in India; therefore, these data are relatively old, which emphasizes the need for countries to strengthen tobacco surveillance, including repeating surveys at regular intervals. Fourth, the current tobacco use was defined as using any tobacco form on at least 1 day in the past 30 days, which might include a proportion of experimenters alongside regular tobacco users. The GYTS questionnaire asked specific questions for cigarette smoking only but did not ask for ST. For example, whether they refused purchase of ST products due to their age, question on ST exposure outside the home or public places, health warning on ST pack was not asked (apart from Bangladesh). Finally, as we wanted to emphasize the wider environmental determinants of tobacco use, we did not include individual-level factors and the exposure at home or parental tobacco use which might have some confounding effect on the observed associations.

Despite these limitations, our study has several notable strengths. The study included a large sample of 23 681 students from four low- and middle-income countries. The GYTS relies on standard procedures for the selection of participants. Nearly the same procedure and questions were applied in all countries, which makes the rationale for merging the data sets to test the hypothesized associations. Most of the previous studies in this region focused on individual-level knowledge, attitude, and family and peer level of behavioral determinants of tobacco use, but not on the wider environmental predictors as explored in our study. Unlike our study, very limited studies have investigated the associations of these environmental-level factors with cigarette smoking, ST, and dual use separately among adolescents.

Although Bangladesh, India, Pakistan, and Sri Lanka are signatory of WHO–FCTC and have country-specific tobacco control laws and policies, tobacco uptake among youth is still high in these countries. Antitobacco policies under FCTC have not been adequately implemented in South Asian countries,[10,43] and social acceptability of ST consumption[44] makes it even harder to implement these comprehensively. Our study reveals that the associations between different forms of tobacco use and protobacco factors were strong, whereas antitobacco factors lacked strength and consistency in the study population. Our study also suggests that the current form of antitobacco media and social campaigns were unlikely to have worked in the study population. Tobacco control strategy should focus on strengthening and enforcing comprehensive bans on smoking in public places, tobacco advertising (including electronic media exposure), sponsorship, and promotions on sale to minors to eliminate protobacco influences in youth.

Supplementary Material

A Contributorship Form detailing each author's specific involvement with this content, as well as any supplementary data, are available online at https://academic.oup.com/ntr.

Acknowledgments

The authors wish to thank the Addressing Smokeless Tobacco and building Research capacity in South Asia (ASTRA) Global Health research team. No ethics approval was required for this study.

This content downloaded from
103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

Funding

This study was supported by grants received from National Institute of Health Research (NIHR) for ASTRA project (GHR Project: 17/63/76).

Author Contribution

KS and MPM conceptualized the study. MK, OD, and KS provided guidance on data analysis. RS conducted the data cleaning and merging for the study. MPM conducted the data analysis and drafted the original manuscript. FS helped draft the manuscript. All authors reviewed and edited the drafts of the manuscript, provided critical feedback. and approved the final manuscript. FS and RS have equal contribution on this article.

Declaration of Interests

None declared.

References

1. National Center for Chronic Disease Prevention and Health Promotion (US) Office on Smoking and Health. *Preventing Tobacco Use Among Youth and Young Adults: A Report of the Surgeon General*. Atlanta, GA: Centers for Disease Control and Prevention (US); 2012. https://www.ncbi.nlm.nih.gov/books/NBK99237/.
2. Xi B, Liang Y, Liu Y, et al. Tobacco use and second-hand smoke exposure in young adolescents aged 12–15 years: data from 68 low-income and middle-income countries. *Lancet Glob Health*. 2016;4(11):e795–e805. doi:10.1016/S2214-109X(16)30187-5
3. ASH. Smoking Statistics: Who Smokes and How Much. 2018. https://ash.org.uk/information-and-resources/fact-sheets/smoking-statistics-who-smokes-and-how-much/.
4. Pierce JP, Sargent JD, Portnoy DB, et al. Association between receptivity to tobacco advertising and progression to tobacco use in youth and young adults in the PATH Study. *JAMA Pediatr*. 2018;172(5):444–451.
5. Pierce JP, Sargent JD, White MM, et al. Receptivity to tobacco advertising and susceptibility to tobacco products. *Pediatrics*. 2017;139(6):e20163353. doi:10.1542/peds.2016–3353
6. Wen CP, Chen T, Tsai Y-Y, et al. Are marketing campaigns in Taiwan by foreign tobacco companies targeting young smokers? *Tob Control*. 2005;14(suppl 1):i38–i44. doi:10.1136/tc.2004.007971
7. Rani M, Thamarangsi T, Agarwal N. Youth tobacco use in South-East Asia: implications for tobacco epidemic and options for its control in the region. *Indian J Public Health*. 2017;61(suppl 1):S12–S17. doi:10.4103/ijph.IJPH_241_17
8. World Health Organization. WHO Report on the Global Tobacco Epidemic, 2017: Monitoring Tobacco Use and Prevention Policies. World Health Organization; 2017. https://apps.who.int/iris/handle/10665/255874.
9. Sinha DN, Kumar A, Bhartiya D, et al. Smokeless tobacco use among adolescents in global perspective. *Nicotine Tob Res*. 2017;19(11):1395–1396.
10. National Cancer Institute, World Health Organization. Monograph 21. The Economics of Tobacco and Tobacco Control. NIH Publ No 16-CA-8029A. 2016. http://cancercontrol.cancer.gov/brp/tcrb/monographs. Accessed December 18, 2019.
11. Anderson CL, Becher H, Winkler V. Tobacco control progress in low and middle income countries in comparison to high income countries. *Int J Environ Res Public Health*. 2016;13(10):1039. doi:10.3390/ijerph13101039
12. Mushtaq N, Huque R, Beebe LA, Shah S, Siddiqi K. Evaluation of tobacco dependence measures in South Asian smokeless tobacco users. *Drug Alcohol Depend*. 2019;203:66–71.
13. Schensul JJ, Nair S, Bilgi S, et al. Availability, accessibility and promotion of smokeless tobacco in a low-income area of Mumbai. *Tob Control*. 2013;22(5):324–330.
14. Kakde S, Bhopal RS, Jones CM. A systematic review on the social context of smokeless tobacco use in the South Asian population: implications for public health. *Public Health*. 2012;126(8):635–645.
15. Wellman RJ, Dugas EN, Dutczak H, et al. Predictors of the onset of cigarette smoking: a systematic review of longitudinal population-based studies in youth. *Am J Prev Med*. 2016;51(5):767–778.
16. Wellman RJ, Sugarman DB, DiFranza JR, Winickoff JP. The extent to which tobacco marketing and tobacco use in films contribute to children's use of tobacco: a meta-analysis. *Arch Pediatr Adolesc Med*. 2006;160(12):1285–1296.
17. Lovato C, Watts A, Stead LF. Impact of tobacco advertising and promotion on increasing adolescent smoking behaviours. *Cochrane Database Syst Rev*. 2011;(10):CD003439. doi:10.1002/14651858.CD003439.pub2
18. Soneji S, Ambrose BK, Lee W, Sargent J, Tanski S. Direct-to-consumer tobacco marketing and its association with tobacco use among adolescents and young adults. *J Adolesc Health*. 2014;55(2):209–215.
19. Leão T, Kunst AE, Schreuders M, Lindfors P, Kuipers MA, Perelman J; SILNE-R Group. Adolescents' smoking environment under weak tobacco control: a mixed methods study for Portugal. *Drug Alcohol Depend*. 2019;204:107566.
20. Dunlop S, Kite J, Grunseit AC, et al. Out of sight and out of mind? Evaluating the impact of point-of-sale tobacco display bans on smoking-related beliefs and behaviors in a sample of Australian adolescents and young adults. *Nicotine Tob Res*. 2015;17(7):761–768.
21. White VM, Durkin SJ, Coomber K, Wakefield MA. What is the role of tobacco control advertising intensity and duration in reducing adolescent smoking prevalence? Findings from 16 years of tobacco control mass media advertising in Australia. *Tob Control*. 2015;24(2):198–204.
22. Thomas RE, McLellan J, Perera R. School-based programmes for preventing smoking. *Cochrane Database Syst Rev*. 2013;(4):CD001293. doi:10.1002/14651858.CD001293.pub3
23. Al-Sadat N, Misau AY, Zarihah Z, Maznah D, Tin Tin Su. Adolescent tobacco use and health in Southeast Asia. *Asia Pac J Public Health*. 2010;22(3 suppl):175S–180S.
24. Rahman MM, Ahmad SA, Karim MJ, Chia HA. Determinants of smoking behaviour among secondary school students in Bangladesh. *J Community Health*. 2011;36(5):831–838.
25. Hussain A, Zaheer S, Shafique K. Individual, social and environmental determinants of smokeless tobacco and betel quid use amongst adolescents of Karachi: a school-based cross-sectional survey. *BMC Public Health*. 2017;17(1):913.
26. Oswal KC. Factors associated with tobacco use among adolescents in India. *Asia Pacific J Public Heal*. 2015;27(2):NP203–NP211. doi:10.1177/1010539511436322
27. Kabir MA, Goh KL, Khan MH. A cross-country comparison of tobacco consumption among youths from selected South-Asian countries. *BMC Public Health*. 2013;13:379.
28. Centre for Disease Control and Prevention (CDC). https://www.cdc.gov/tobacco/global/index.htm. Accessed February 10, 2019.
29. Warren CW, Lee J, Lea V, et al. Evolution of the Global Tobacco Surveillance System (GTSS) 1998–2008. *Glob Heal Promot*. 2009;16(2 suppl):4–37. doi:10.1177/1757975909342181
30. GYTS Pakistan. Global Youth Tobacco Survey (GYTS) Fact Sheet Pakistan. 2013. http://www.emro.who.int/images/stories/tfi/documents/GYTS_FS_PAK_2013.pdf?ua=1. Accessed December 22, 2019.
31. GYTS Sri Lanka. Global Youth Tobacco Survey (GYTS) Fact Sheet Sri Lanka. 2015. http://origin.searo.who.int/tobacco/data/gyts_sri_lanka_2015_factsheet.pdf. Accessed December 22, 2019.
32. GYTS Bangladesh. Global Youth Tobacco Survey (GYTS) Fact Sheet Bangladesh. 2013. http://origin.searo.who.int/tobacco/data/ban_gyts_factsheet_2013.pdf?ua=1. Accessed December 22, 2019.
33. GYTS India. Global Youth Tobacco Survey (GYTS) Fact Sheet India. 2009. https://www.who.int/fctc/reporting/Annexoneindia.pdf. Accessed December 22, 2019.
34. Gajalakshmi V, Kanimozhi C V. A survey of 24,000 students aged 13–15 years in India: Global Youth Tobacco Survey 2006 and 2009. *Tob Use Insights*. 2010;3:23–31. doi:10.1177/1179173X1000300001
35. StataCorp LLC. Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC; 2019.
36. Mehrotra R, Yadav A, Sinha DN, et al. Smokeless tobacco control in 180 countries across the globe: call to action for full implementation of WHO FCTC measures. *Lancet Oncol*. 2019;20(4):e208–e217.

This content downloaded from
103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms

37. Warren CW, Jones NR, Eriksen MP, Asma S; Global Tobacco Surveillance System (GTSS) Collaborative Group. Patterns of global tobacco use in young people and implications for future chronic disease burden in adults. *Lancet*. 2006;367(9512):749–753.
38. Rudatsikira E, Muula AS, Siziya S. Current use of smokeless tobacco among adolescents in the Republic of Congo. *BMC Public Health*. 2010;10:16.
39. Gilmore AB, Fooks G, Drope J, Bialous SA, Jackson RR. Exposing and addressing tobacco industry conduct in low-income and middle-income countries. *Lancet*. 2015;385(9972):1029–1043.
40. Shroff H, Surani N. Mass media exposure to tobacco messages among secondary school children in Mumbai. *Indian J Public Health*. 2012;56(2):159. doi:10.4103/0019-557X.99917
41. Nonnemaker J, Farrelly M, Kamyab K, Busey A, Mann N. Experimental Study of Graphic Cigarette Warning Labels: Final Results Report. Research Triangle Park, NC: RTI International, United States; 2010.
42. Pepper JK, Cameron LD, Reiter PL, McRee AL, Brewer NT. Non-smoking male adolescents' reactions to cigarette warnings. *PLoS One*. 2013;8(8):e65533.
43. Mamudu H, Cairney P, Studlar D. Global public policy: does the new venue for transitional tobacco control challenge the old way of doing things? *Public Adm*. 2016;93(4):856–873. doi:10.1111/padm.12143
44. Anwar S, Williams SA, Scott-Smith J, et al. A comparison of attitudes and practices of gutka users and non-users in Chitrakoot, India. A pilot. *Prim Dent Care*. 2005;12(1):5–10.

This content downloaded from
103.99.176.50 on Thu, 07 May 2026 15:23:52 UTC
All use subject to https://about.jstor.org/terms